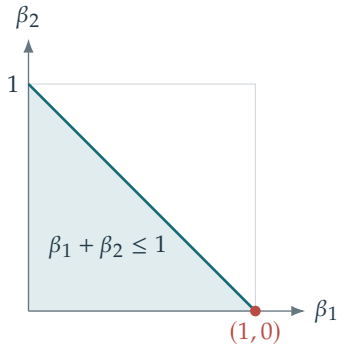
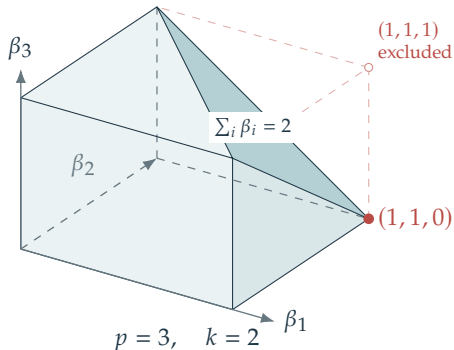




$$p = 2, \quad k = 1$$



Feasible region: $0 \leq \beta_i \leq 1, \sum_i \beta_i \leq k$. Red dots: a maximizing vertex.

$$\lambda_1 \geq \cdots \geq \lambda_p \geq 0, \quad \max \sum_i \lambda_i \beta_i = \sum_{i=1}^k \lambda_i$$